

Entropy is NOT All You Need: Mechanistic Effects of Attention Inverse-Temperature in Transformers

Relationship to Entropic-Brain and REBUS Accounts of Psychedelics

Introduction

Overview

This project examines how attention inverse-temperature (β) mechanistically shapes the generative behavior of a transformer (Llama-3.2-1B-Instruct), motivated by an analogy to neuroscientific accounts of the psychedelic state. Under the modern Hopfield interpretation of attention, retrieval can be understood as settling into minima of an energy landscape; lowering β flattens that landscape and blurs attention sharpness so that retrieval becomes less peaked. This operation is conceptually analogous to a central motif in the psychedelic literature: REBUS, short for Relaxed Beliefs Under pSychedelics, (Carhart-Harris & Friston, 2019) describes the hypothesized flattening of the brain's variational free-energy landscape under psychedelics as analogous to simulated annealing. While these two landscapes are distinct objects, with REBUS operating over hierarchical beliefs and β acting on a Hopfield energy landscape over attention states, both frameworks invoke the same landscape-flattening picture. Because β is an inverse-temperature parameter, lowering it implements the same temperature-raising move in attention space that REBUS gestures at with the annealing analogy. Thus, we predict that flattening the attention energy landscape via β would causally produce loosened, psychedelic-like reasoning, reflecting the entropic brain hypothesis (EBH) in which raised disorder drives the altered state (Carhart-Harris et al., 2014).

Methods

Model and runtime

All experiments use `meta-llama/Llama-3.2-1B-Instruct` (16 transformer layers), loaded through HuggingFace Transformers with `attn_implementation="eager"`. The model runs in float32 on CPU/MPS and float16 on CUDA with weights frozen such that every intervention acted at inference time by patching the attention computation.

The β Attention Inverse-Temperature Intervention

Attention computes `softmax(QKT · scaling)`, where `scaling = 1/√d =  $\beta_0$` is the native inverse-temperature. We patch the eager attention function such that on a selected set of layers, `scaling` is multiplied by a ratio `r =  $\beta/\beta_0 \in (0, 1]$` read live from a global `InterventionState` on every forward pass. Setting `r < 1` lowers the effective inverse-temperature (flattening the attention distribution and the Hopfield energy landscape) and `r = 1` leaves attention untouched. Upon a

preliminary parameter sweep, we determined that the most effective intervention to induce psychedelic reasoning was $r = 0.45$ applied on layers 2–3.

Stimuli (prompt set)

Reasoning is probed with open-ended generation rather than fixed-answer tasks because the hypothesis centers around associative loosening, which closed tasks cannot reveal. A fixed-seed pool of open prompts is drawn in balanced proportion from four types: (i) continuation of an evocative passage; (ii) free-association from a seed word; (iii) vivid description of a scene; and (iv) loose, reflective Q&A. Experiments use $n = 8$ prompts for the β screens with greedy generation (`do_sample=False`) unless otherwise noted, so that measured differences are attributable to the intervention rather than sampling stochasticity.

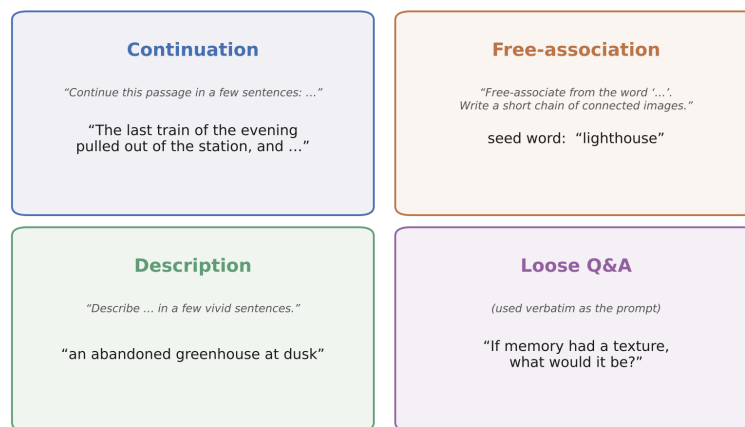


Figure 1: The four open-ended prompt pools, with one example seed each. Prompts were sampled from these pools (28 seeds total) using a fixed seed.

Outcome metrics

Upon probing the model, we computed two behavioral metrics under a clean (un-intervened) model state to establish a rudimentary evaluation of psychedelic reasoning.

- **Associative drift** = $1 - \text{cosine_similarity}(\mathbf{e_prompt}, \mathbf{e_output})$, where each embedding is the mean-pooled last-layer hidden state of the text under the clean model. Higher drift indicated that the output wandered further from the prompt.
- **Perplexity** = $\exp(\text{mean token NLL})$ of the generated text under the clean model. Lower perplexity indicated higher local coherence, calculated as $1 / (1 + \max(0, \log(\text{ppl}) - \log(10)))$.

When considered together, the two axes separate interesting loosening (high drift, coherence preserved) from mere degradation (coherence collapse).

Entropy / complexity measures (EBH candidates)

To evaluate our hypothesis, we measured four entropy indices as candidate analogues of the entropic-brain signature, under β flattening.

1. **Attention-weight entropy.** Normalized Shannon entropy of the attention weight distribution, $H = -\sum w_i \log w_i / \log k$, captured per layer via a lightweight forward hook on the target layers' attention modules.
2. **Residual-stream Lempel-Ziv complexity (LZc).** Each unit's activation trajectory across tokens is binarized at its per-unit median, concatenated, and compressed with a Kaspar-Schuster LZ76 parse, normalised by $n / \log_2 n$. This serves as an analogue of the EEG/MEG measure, which is shown to increase under the presence of psychedelics (Schartner et al., 2017).
3. **Residual-stream spatial complexity.** Per token, the Shannon entropy of the normalised distribution of `|activation|` across hidden units, normalised by `log(units)` and averaged over tokens. Shown to increase under the presence of psychedelics (Siegel et al., 2024).
4. **Attention-graph degree-distribution entropy.** Viol et al. (2017) found that under ayahuasca, the brain's functional-connectivity network shows increased Shannon entropy of its degree distribution, indicating a more heterogeneous, hub-rich connectivity structure. We port this measure to the transformer by treating the attention matrix as a token–token connectivity graph — heads are averaged, the matrix is symmetrized $(A + A^T)/2$ and zero-diagonalized, then thresholded to a target edge density (top-k strongest edges). Then we simply computed the Shannon entropy of the resulting degree distribution.

Causal dissociation

Upon analysis of the EBH measures, we determined that the β -intervention most directly changed the attention-weight entropy. However, because the attention output is a weighted sum of value vectors, β does not only flatten the attention weights — it also changes the resulting weighted sum, altering the representation passed to the next layer.

In order to cleanly demonstrate whether the increase in attention-weight entropy *causally* altered reasoning, we ran a paired sufficiency-necessity intervention to decouple the effects of entropy and the attention output.

1. **Sufficiency** – A hook on the `o_proj` input perturbs the attention output directly, while leaving the attention weights and hence their entropy at baseline, isolating the contribution of the resultant weighted sum itself. The fraction of β 's behavioural effect produced by this intervention measures whether changing the output alone is sufficient to drive the effect.
2. **Necessity** – A custom attention replacement computes the baseline weights `w` and the flattened weights `w_flat = (1 - flatten) · w + flatten · (1/k)`, then writes `out_final = out_flat + restore · (out_base - out_flat)`. This raises attention-weight entropy while holding the output near baseline, isolating the entropy pathway. The fraction of β 's effect produced by this intervention measures whether attention-weight entropy is necessary to alter reasoning.

Direct degree-entropy intervention

Interestingly, we noticed that the attention-graph degree-entropy *decreased* under β flattening, contradicting the findings in the original Viol et al. (2017) research, prompting us to run a secondary intervention independent of β to determine its effects on reasoning. In order to raise the degree-entropy, we implemented a custom attention replacement which reweights the decode-step attention logits by adding a smooth per-key hubness bias ($\text{bias} = \text{strength} \cdot 3.0 \cdot \cos(\pi k / (K-1))$) to push the thresholded graph toward an uneven, hub-rich degree distribution. Like with the β -intervention, we ran a layer-sweep to determine if specific target layers produced greater shifts in reasoning; however, since this was not the case, we applied the intervention to each of the 16 layers. For this experiment, we chose to screen with $n = 6$ prompts with greedy generation.

Simulated-annealing β schedule

To test the cooling half of the REBUS energy landscape annealing analogy, we implemented a dynamic schedule in which β is varied *within* a single generation rather than held fixed. Because the standard generate loop fixes β for the whole call, generation is run as a manual token-by-token greedy decode loop that mutates `INTERVENTION.beta_ratio` before each step according to a schedule $f(t)$ over generation progress $t \in [0, 1]$. Schedules interpolate between a hot end $\beta_{\text{lo}} = 0.45$ and a cold end $\beta_{\text{hi}} = 1.0$ and are implemented through linear ($\text{lo} + (\text{hi}-\text{lo}) \cdot t$), exponential ($\text{lo} + (\text{hi}-\text{lo}) \cdot (1 - e^{(-3t)}) / (1 - e^{(-3)})$), quadratic ($\text{lo} + (\text{hi}-\text{lo}) \cdot t^2$), and square root ($\text{lo} + (\text{hi}-\text{lo}) \cdot \sqrt{t}$). Each schedule's mean drift and coherence are compared against a grid of static β values (0.45–1.0) in drift $\times$ coherence space. The decisive test is whether any schedule achieves higher drift than the best static β at matched coherence, in order to determine whether the model produced more “wandering” without losing coherence. For this experiment, we also chose to screen with $n = 6$ prompts.

Results

Flattening β to 0.45 on layers 2–3 substantially loosened generation while leaving it non-degenerate. Local coherence collapsed, with perplexity rising from ≈ 3.6 at baseline to ≈ 17 – 19 under β (and as high as ≈ 26 when layer 3 was flattened alone) and the bounded coherence score fell from 1.0 to ≈ 0.79 . Associative drift moved in the loosening direction (≈ 0.30 to ≈ 0.21), consistent with a genuinely “loosened” generative regime rather than simple breakage. This confirms β is an active, behaviorally consequential intervention and establishes the effect whose cause the remainder of the analysis isolates.

Measured against the four candidate entropy indices, β 's effect did *not* present as a uniform increase in disorder. Attention-weight entropy rose at the intervened layers ($\approx +0.3$ in normalized entropy at layers 2 and 3), consistent with the EBH prediction. But the residual-stream complexity measures did not follow: Lempel-Ziv complexity slightly *decreased* (≈ 4 – 8% , target-layer and network-mean both negative), and spatial complexity was essentially flat (≈ 0 – 0.16% change). Most strikingly, the ported Viol et al. (2017) measure moved in the *opposite* direction to the psychedelic signature: attention-graph degree-distribution entropy *decreased* by ≈ 0.51 nats under β , accompanied by reduced clustering and roughly flat global

efficiency. Whereas ayahuasca raises degree entropy and local integration, β drove the attention connectivity toward a more regular, lower-entropy structure. Raising degree-entropy via the hubness reweighting intervention succeeded mechanically, achieving increases of $\approx +0.2$ to $+0.43$ nats. However, at every layer, raising degree entropy left reasoning essentially unchanged: coherence remained at 1.0 and perplexity stayed near baseline (≈ 4.0 – 4.7 vs. ≈ 4.1), with drift shifts within noise (± 0.02). Because this intervention produced no loosening at any layer, we were able to test the causal claim that correlational neuroimaging cannot.

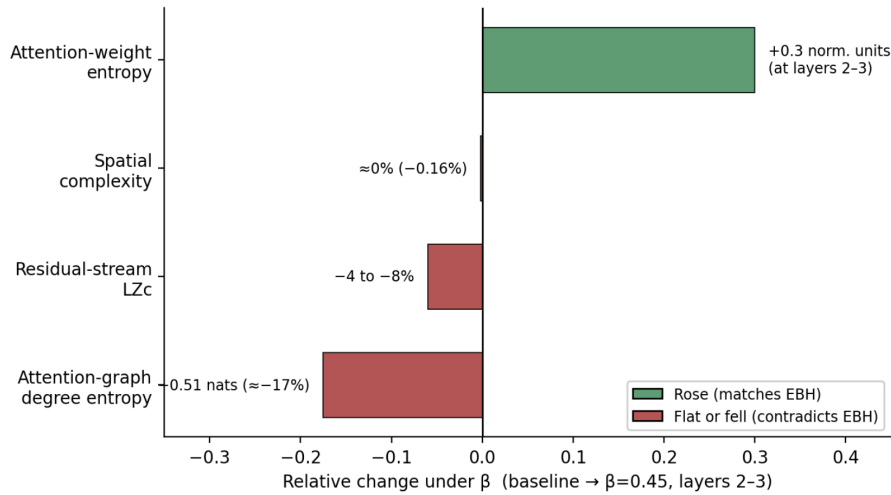


Figure 2: Only one of four entropy measures, the attention-weight entropy, rose under β , while the degree-entropy measure moved opposite to the psychedelic (EBH) prediction.

The bulk of our work centered upon the results of the paired sufficiency-necessity intervention to determine whether the attention weight entropy itself or the attention output (the weighted sum of value vectors) was responsible for the shift in reasoning. The **sufficiency** test revealed that a structured perturbation reproduced $\approx 76\%$ of β 's drift effect, meaning that changing the output alone is therefore largely sufficient to reproduce β 's effect. The **necessity** test did the converse, yet despite the large entropy increase and a non-trivial residual output shift of ≈ 0.59 , this reproduced only $\approx 5\%$ of β 's effect (perplexity fraction 0.053, drift fraction 0.054), with coherence remaining at 1.0. In other words, together these establish that the attention output is sufficient ($\approx 76\%$) and weight entropy is not necessary ($\approx 5\%$).

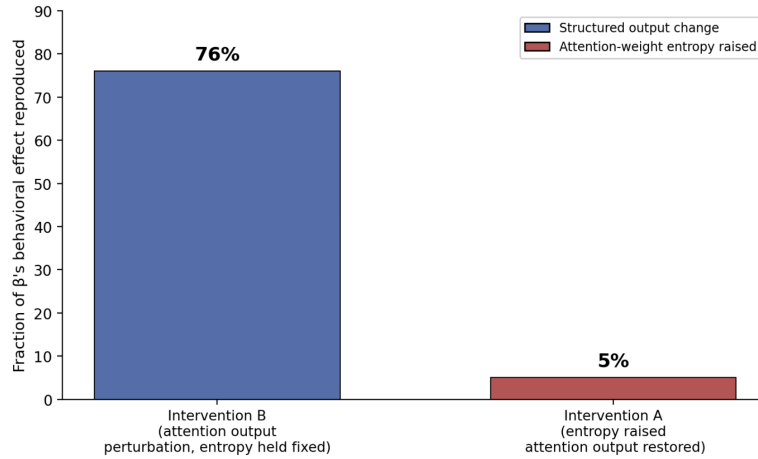


Figure 3: Perturbing the attention output (entropy fixed) reproduces $\approx 76\%$ of β 's drift effect, while raising attention-weight entropy (output restored) reproduces only $\approx 5\%$. Thus, the attention output is sufficient and weight entropy is not necessary.

Finally, we tested whether scheduling β over generation could reach a state of high drift with preserved coherence that no static β achieves. The static- β frontier established that the highest drift achievable at full coherence (1.0) was ≈ 0.27 at $\beta = 0.75$. However, every schedule that raised drift above this bar did so only by sacrificing coherence: the square-root and exponential schedules reached drift ≈ 0.30 – 0.32 but at coherence ≈ 0.71 (perplexity ≈ 22), the quadratic schedule reached drift ≈ 0.30 at coherence ≈ 0.88 (perplexity ≈ 11), and the linear schedule landed at drift ≈ 0.26 at coherence ≈ 0.93 . Rather than escaping the drift–coherence trade-off onto a new frontier, the schedules re-established the same trade-off that lowering static β already traces.

Discussion

Interpretation

Our central finding is a dissociation between signature and cause. Flattening β loosened generation and raised some of the entropy signatures EBH associates with the psychedelic state, such as attention-weight entropy. However, the interventions establish that raising attention-weight entropy while holding the attention output fixed reproduced $\approx 5\%$ of the effect and perturbing the output while holding entropy fixed reproduced $\approx 76\%$. Entropy is therefore a co-occurring readout of the reshaped attention weights, while the actual mechanism that carries the effect is the structured change to the attention output as it propagates through the residual stream. This recapitulates, in an artificial system, the relationship between the two dominant accounts of psychedelics. EBH characterizes the altered state by elevated entropy (Carhart-Harris et al., 2014) while REBUS characterizes the mechanism as relaxed high-level priors that change how signals are weighted and combined (Carhart-Harris & Friston, 2019). Our entropy indices behaved as signatures, while the causal lever was re-weighted signal integration — reduced prior precision in REBUS and a structured change to the attention-weighted value vector sum in the transformer.

The degree-entropy result makes this sharpest by turning a correlational neuroscience claim into a causal test. Viol et al. (2017) observed elevated degree-distribution entropy under ayahuasca but was unable to

isolate causation in a human brain. To run this test, we implemented an intervention to raise degree-entropy directly in a transformer which left reasoning effectively unchanged. This confirms that degree-entropy is a marker, not a cause. In fact, under β -flattening, the measure moved the *opposite* way in comparison to neuroscientific literature, so even the correlation does not transfer cleanly. The failed annealing schedule bounds the landscape-flattening picture similarly: scheduling β from hot to cold within a generation did not reach a "lucid but loose" state, but only retraced the static drift-coherence trade-off; likely because autoregressive generation, unlike an optimizer, cannot revise tokens already emitted while hot.

In total, these argue that the entropic framing identifies signatures that co-occur with the loosened state but not its cause. Loosened reasoning here is a consequence of structured changes to how value information is weighted and integrated across depth, with entropy riding along as a readout. The mechanism is real and reproducible, while its description as an entropic phenomenon is not. The analogy to psychedelic cognition, to the extent it holds, holds through REBUS-style re-weighting rather than EBH-style entropy, allowing us to conclude that entropy is *not* all you need.

Limitations

Our study has several important limitations.

- The study is small and single-system: one 1B-parameter model, a fixed primary intervention ($\beta = 0.45$ on layers 2–3), and $n = 6$ –8 prompts per screen with greedy decoding. Results are therefore directional rather than statistically powered, reported as means without modeling per-prompt variance, so the dissociation percentages ($\approx 76\%$ sufficiency, $\approx 5\%$ necessity) should be read qualitatively rather than precisely.
- The brain-transformer analogy is structural, not mechanistic. The attention-graph we use to port Viol et al. (2017) is directed, causally masked, and task-driven, whereas the brain's functional-connectivity graph is undirected, resting-state, and correlation-based. The null result for direct degree-entropy manipulation may therefore reflect either a genuine absence of a causal role for degree-entropy *or* a failure of the analogy to transfer between such different substrates. Similarly, the alignment we draw with REBUS is at the level of mechanism-form (re-weighted signal integration) rather than detail. Crucially, β acts bottom-up at early layers, whereas REBUS specifically implicates the relaxation of high-level priors, so the correspondence should not be read as a claim that the transformer directly implements the predictive-coding hierarchy REBUS presupposes.
- Our operationalization of reasoning is deliberately rudimentary. Associative drift and clean perplexity capture only two axes, semantic wandering and local coherence, without considering other characteristics of psychedelic cognition, such as synesthetic or cross-domain blending, ego dissolution, altered salience and meaning attribution, recursive or self-referential thought, and distortions of time and narrative structure. Our null and dissociation results therefore speak to the specific axes we measured, and a richer operationalization might reveal characteristics of psychedelic-like reasoning, or causal relationships to entropy, that this study could not.

Future work

Several directions follow naturally from these results. The most direct is scaling: repeating the dissociation across larger models, more layers, and adequately powered prompt sets would test whether the value-output mechanism we isolate here generalizes or is specific to a 1B model. A second direction is to enrich the operationalization of "psychedelic-like" reasoning beyond drift and coherence, such as with the metrics defined above, to check whether any axis we did not capture *does* track an entropy measure. Finally, to understand the failed annealing result, we will compare cooling schedules against a revision-capable (e.g. diffusion-style) decoder which would isolate whether the inability to reach a lucid-but-loose state is truly a consequence of irreversible token emission, or merely of average β .

References

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